

Bubble Sort:

Bubble Sort is a simple sorting algorithm that repeatedly compares adjacent elements and swaps them if they are in the wrong order. This process continues until the array is sorted.

Algorithm:

BubbleSort(arr)

1. Start
2. For $i = 0$ to $n - 2$
 - a. For $j = 0$ to $n - i - 2$
 - If $\text{arr}[j] > \text{arr}[j + 1]$
 - Swap $\text{arr}[j]$ and $\text{arr}[j + 1]$
3. Stop

Output:

```
Enter the number of elements: 6
Enter the elements:
8
3
12
45
13
4

Sorted Array:
[3, 4, 8, 12, 13, 45]

Execution Time: 0.0000729560 seconds

Time Complexity:
Best Case      : O(n)
Average Case   : O(n2)
Worst Case     : O(n2)
Space Complexity: O(1)

...Program finished with exit code 0
Press ENTER to exit console.█
```

Time complexity:

- **Best Case: O(n)**

The array is already sorted (with an optimized Bubble Sort that stops early if no swaps occur).

- **Average Case: O(n²)**

Multiple comparisons and swaps are required.

- **Worst Case: O(n²)**

The array is in reverse order, requiring the maximum number of comparisons and swaps.

Space Complexity:

- **Iterative Bubble Sort: $O(1)$**

Uses only a temporary variable for swapping elements.